

3 Cholera is an infectious disease caused by the bacterial pathogen *Vibrio cholerae*.

(a) Describe how *V. cholerae* is transmitted from an infected person to an uninfected person.

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V. cholerae releases a toxin called cholera toxin. Cholera toxin is a protein molecule made up of six polypeptides (subunits).

(b) Fig. 3.1 is a ribbon diagram of a molecule of cholera toxin.



Fig. 3.1

Describe what is shown by the part of the molecule labelled T.

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(c) Vaccination can be used to prevent the spread of cholera. The vaccine used contains part of the cholera toxin molecule and also contains dead cells of *V. cholerae*.

(i) Phagocytes play an important role in the immune response to a vaccine. Phagocytes contain many lysosomes.

Describe the function of lysosomes in phagocytes.

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(ii) Suggest the advantage of having a vaccine that includes dead cells of *V. cholerae* and part of the cholera toxin molecule.

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- (d) The World Health Organization states that most cases of cholera can be treated with oral rehydration therapy. To reduce the risk of antibiotic resistance developing, antibiotics are only recommended for the most serious cases.

Discuss the consequences of the development of antibiotic resistance in *V. cholerae*.

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Question 4 starts on page 12.